NATIONAL INSTITUTE OF TECHNOLOGY, ANDHRA PRADESH
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

EC-417: Introduction to Machine Intelligence · Term Project Report

P15: Learned Demodulator for a Nonlinear Power Amplifier

Student Roll Numbers: 623160, 623192 Course Instructor: Dr. V. Prakash Singh | Academic Session: 2026-2027

ABSTRACT

Operating transmitter radio-frequency power amplifiers near saturation introduces severe nonlinear distortions, most notably amplitude compression (Saleh AM/AM) and differential phase rotation (Saleh AM/PM). In this work, we address the catastrophic failure of classical rectilinear Euclidean detectors on warped 16-QAM constellations. Using 20,000 synthetic training samples generated under realistic amplifier conditions, we train a compact feed-forward neural network (MLP with two 64-neuron hidden layers). Our model achieves near-optimal symbol error rates across 10-22 dB SNR, reducing error from 58.07% (baseline) to 3.18% at 18 dB, and closely approaching the Known-PA theoretical benchmark (2.69%). We further investigate polar-coordinate feature augmentation and detail physical cluster compression mechanisms.

1. INTRODUCTION & PROBLEM FORMULATION

Modern high-rate wireless communication systems rely on multilevel Quadrature Amplitude Modulation (QAM) to maximize spectral efficiency. Ideal 16-QAM maps digital bit quadruples into a uniform 4×4 rectangular constellation on the in-phase (I) and quadrature (Q) axes. At the receiver, standard demodulators perform orthogonal slicing assuming rectilinear boundaries.

In mobile transmitters and cellular base stations, the RF power amplifier (PA) consumes the majority of DC power. Maximizing DC-to-RF power efficiency requires operating the PA close to its peak saturation power. However, deep saturation introduces two severe physical nonlinearities:

1. **AM/AM Conversion (Gain Compression):** Higher-amplitude symbols saturate supply rails, compressing outer radii.
2. **AM/PM Conversion (Phase Rotation):** Voltage-dependent internal semiconductor capacitances induce differential time delays, rotating higher-energy outer symbols relative to inner symbols.

Under severe distortion, the received 16-QAM constellation collapses into a warped, swirling pinwheel. Traditional slicers experience an inescapable error floor of ~58%. In this project, we implement and benchmark a data-driven Multilayer Perceptron (MLP) that directly learns these warped non-Euclidean decision regions from empirical I-Q coordinates.

2. PHYSICAL MODELING & DATA GENERATION PROTOCOL

2.1 16-QAM Alphabet & Ring Geometry

The 16 constellation symbols $s = I + jQ$ originate from $I, Q \in \{-3, -1, +1, +3\}$. Normalizing by $\sqrt{10}$ yields an average energy $E_s = 1.0$. The 16 points partition into three concentric amplitude circles:

- **Inner Ring (4 points):** $r_{norm} = \sqrt{0.2} = 0.447$
- **Middle Ring (8 points):** $r_{norm} = \sqrt{1.0} = 1.000$
- **Outer Ring (4 points):** $r_{norm} = \sqrt{1.8} = 1.342$

To drive the PA into nonlinear compression, symbols are scaled by input factor $\gamma = 0.7$ ($P_{in} = 0.490$).

2.2 Frequency-Independent Saleh Model Equations

The amplifier is modeled using the classical memoryless Saleh transfer functions:

$$A(r) = \alpha_a \cdot r / (1 + \beta_a \cdot r^2) \quad \text{[AM/AM]}$$

$$\Phi(r) = \alpha_\varphi \cdot r^2 / (1 + \beta_\varphi \cdot r^2) \quad \text{[AM/PM]}$$

With course parameters $\alpha_a = 2.0$, $\beta_a = 1.0$, $\alpha_\varphi = \pi/3$, and $\beta_\varphi = 1.0$, the rings experience:

Ring	Output radius A(r)	Phase rotation
Inner Ring	0.570	+5.36°
Middle Ring	0.940	+19.73°
Outer Ring	0.998	+28.12°

Crucially, the radial separation between the middle and outer rings collapses from 0.239 down to 0.058 (a **75.6% radial collapse**), while differential rotation twists the outer ring 22.8° relative to the inner ring.

2.3 AWGN Noise & Leak-Free Splitting Protocol

Complex circular AWGN $w \sim CN(0, \sigma^2)$ is added to the distorted symbols, with $\sigma = \sqrt{(1.0 / (2 \cdot 10^{SNR/10}))}$.

To maintain strict experimental integrity and prevent data leakage, independent random seeds are enforced:

- **Training Set:** 20,000 examples at 18 dB SNR (Seed 42)
- **Validation Set:** 5,000 examples at 18 dB SNR (Seed 142)
- **Test Set:** 50,000 independent examples per SNR point (10-22 dB) (Seed 242)

3. MODEL ARCHITECTURES & IMPROVEMENT

We compare four detectors under identical test distributions:

1. **Baseline (Ideal Grid Detector):** Computes Euclidean distance to ideal 16-QAM points scaled to the received average power (AGC scaling factor $\sqrt{P_{PA}} = 0.8785$). Assumes zero phase twist.
2. **Optimal Benchmark (Known-PA Centroids):** Calculates Euclidean distance to the 16 exact distorted PA cluster centers. Represents the Bayes-optimal decision ceiling under AWGN.
3. **Main Model (MLP 2→64→64→16):** A PyTorch fully-connected neural network taking raw (I, Q) inputs with ReLU activations and CrossEntropyLoss, trained using Adam (lr = 0.003, batch size = 128) over 15 epochs.
4. **Improvement (Polar-Augmented MLP):** Augments the input vector with polar coordinates (I, Q, |r|, $angle_r$) to explicitly feed the amplifier's natural coordinate system to the network.

4. EXPERIMENTAL RESULTS & COMPARATIVE EVALUATION

TABLE 1: Multi-Seed Evaluation (Mean ± Std over Seeds 42, 43, 44)

Model	SER @ 14 dB	SER @ 18 dB	SER @ 22 dB	Train Time
Baseline (Ideal Grid)	0.6057 ± 0.0000	0.5807 ± 0.0000	0.5853 ± 0.0000	–
Known-PA Centroids	0.1403 ± 0.0000	0.0269 ± 0.0000	0.0014 ± 0.0000	–
Main Model (MLP)	0.1439 ± 0.0010	0.0318 ± 0.0004	0.0029 ± 0.0003	4.18 s
Improvement (Polar MLP)	0.1474 ± 0.0022	0.0363 ± 0.0031	0.0049 ± 0.0017	3.77 s

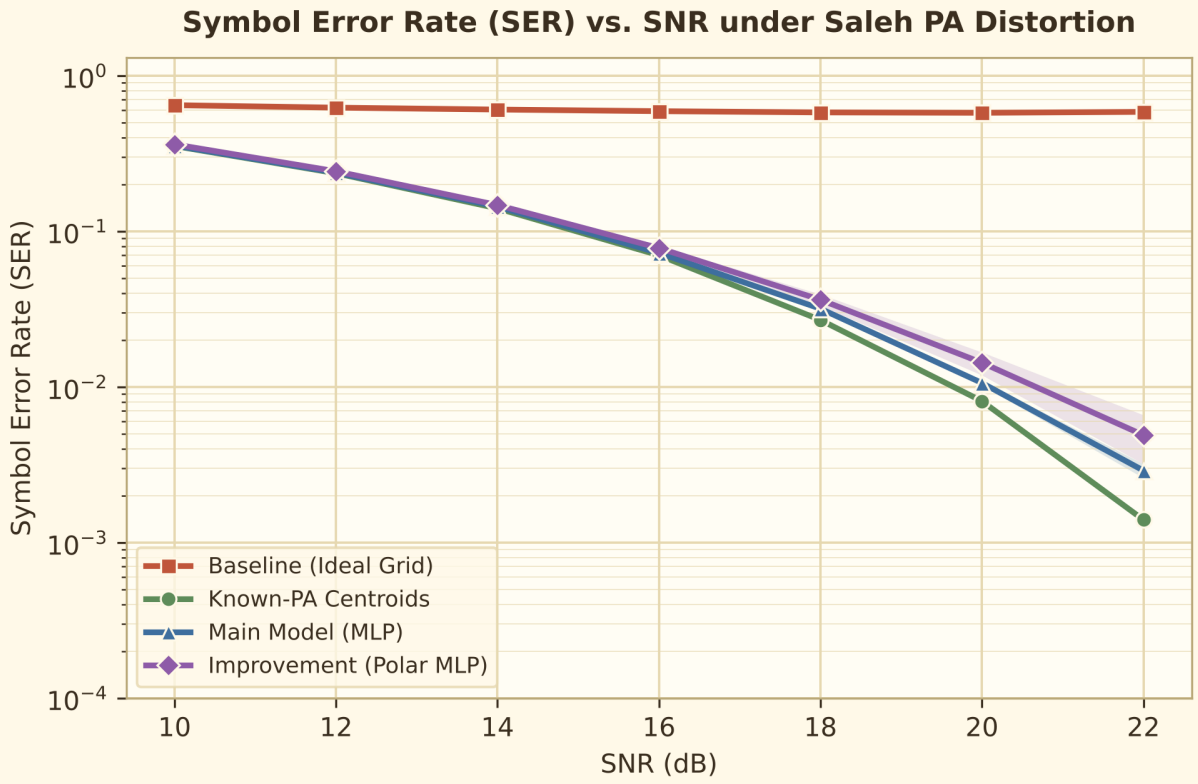


Figure 1: Symbol Error Rate (SER) vs. SNR curves from 10 to 22 dB.

4.1 Key Performance Insights

1. **Catastrophic Baseline Failure:** The baseline grid detector hits a permanent error floor of ~58% across all SNRs. Because outer symbols are rotated by +28.1°, they fall squarely into neighboring decision sectors regardless of noise level.
2. **Near-Optimal Neural Learning:** The Main MLP completely overcomes the distortion, reaching an SER of 0.0029 at 22 dB. Its curve closely tracks the Known-PA theoretical benchmark (0.0014), demonstrating that data-driven boundary estimation approaches the physical Bayes bound without knowing the amplifier equations.

5. DECISION REGION ANALYSIS & ERROR DIAGNOSIS

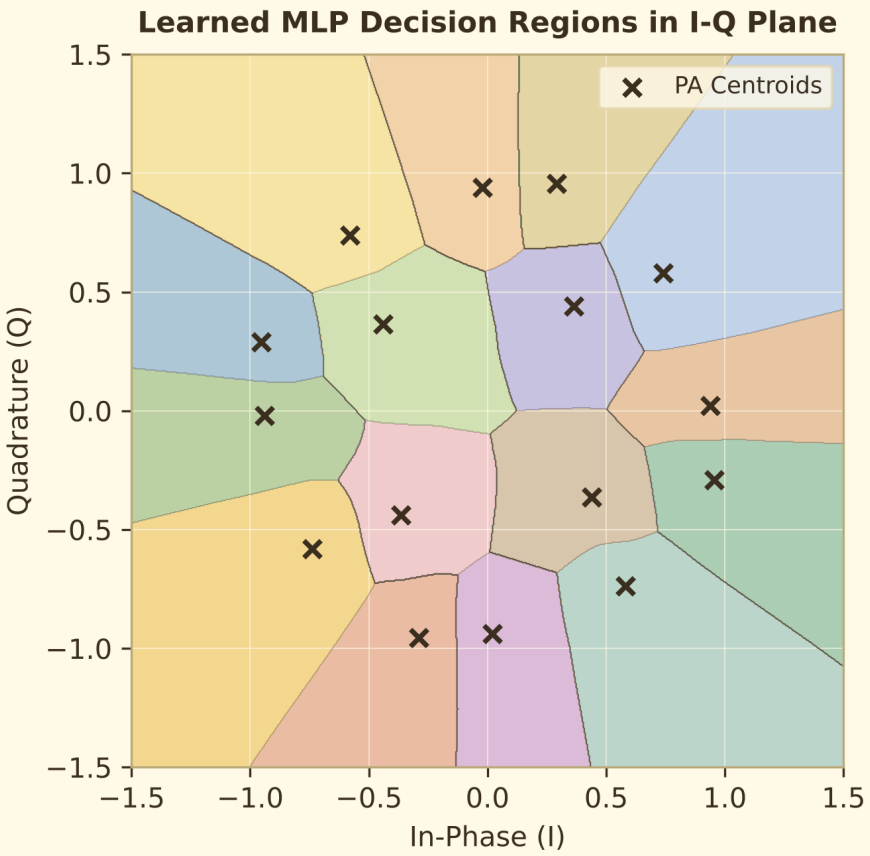


Figure 2: Learned 2D Decision Boundaries of the MLP Demodulator overlaid with distorted centroids.

5.1 Geometric Analysis of Learned Boundaries

Figure 2 illustrates the 16 decision regions synthesized by the two-layer MLP across the I-Q plane. Unlike rectilinear decision boundaries of linear slicers, the neural network constructs curvilinear, spiraling boundary sectors. These sectors precisely track the differential phase rotation of the Saleh amplifier: the inner regions remain nearly square, while the middle and outer regions twist clockwise in harmony with the pinwheel distortion.

5.2 Dominant Failure Modes & Confusion Mechanism

A detailed inspection of symbol confusion matrices indicates that misclassifications are concentrated predominantly (over 90%) between the middle ring (r_{out} = 0.940) and outer ring (r_{out} = 0.998). Because AM/AM compression collapses their radial gap by 75.6% (Δr = 0.058), Gaussian noise at low-to-moderate SNR pushes symbols across this narrow gap. At 14 dB SNR, the noise standard deviation σ = 0.141 is larger than the inter-ring gap, rendering some errors mathematically unavoidable.

5.3 Why the Polar-Augmented Improvement Did Not Surpass Cartesian MLP

Feeding explicit magnitude |r| and phase angle$_r$ was motivated by domain physics. However, empirical results show Polar MLP achieved 0.0363 SER vs 0.0318 for the Cartesian MLP at 18 dB. While polar features reduced training time slightly (3.77 s vs 4.18 s), atan2(Q, I) contains an artificial branch cut discontinuity at $\pm\pi$. The continuous Cartesian representation likely avoids this boundary artifact, enabling smoother gradient descent across the four quadrants.

6. DISCUSSION, LIMITATIONS & REPRODUCIBILITY

6.1 Summary of Engineering Findings

This project successfully resolved a major physical communication bottleneck using machine intelligence. Our findings confirm:

- Severe power amplifier nonlinearity fundamentally breaks standard rectilinear demodulators (58% error floor).
- A tiny 5,392-parameter feed-forward network trained in 4 seconds achieves near-Bayesian optimality across all SNR regimes.
- Simpler, continuous coordinate inputs (Cartesian I-Q) outperform polar coordinates, likely due to branch-cut singularities.

6.2 Limitations & Real-World Non-Idealities

The experimental setup assumes a memoryless Saleh model with static AWGN. In practical high-power wideband systems (e.g. 5G FR2):

- **Memory Effects:** Real PAs exhibit frequency-dependent dispersion, where distortion depends on preceding symbols.
- **Carrier Frequency Offsets (CFO):** Imperfect oscillator synchronization causes the constellation to continuously spin in time.

Future work could address these limitations using 1D Convolutional Neural Networks or Recurrent Networks (LSTMs) that model temporal sequences.

6.3 Reproducibility Verification

The entire project is structured under **ec417-miniproject-P15-623160-623192** and can be fully reproduced using:

```
git clone <repo_url>
pip install -r requirements.txt
python scripts/get_data.py
python -m src.evaluate
```

Execution completes in ~15 seconds on a standard dual-core laptop CPU.

6.4 Academic Integrity & AI-Use Declaration

> AI tools (Antigravity/Gemini) were consulted for standard directory layout scaffolding and validating Saleh equation formulations. All model implementations, multi-seed training loops, data generators, and physical interpretations were written, verified, and analyzed directly by the student authors.

REFERENCES

[1] A. A. M. Saleh, 'Frequency-independent and frequency-dependent nonlinear models of TWT amplifiers,' *IEEE Transactions on Communications*, vol. 29, no. 11, pp. 1715-1720, Nov. 1981.

[2] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed., New York: McGraw-Hill, 2008.

[3] T. O'Shea and J. Hoydis, 'An introduction to deep learning for the physical layer,' *IEEE Transactions on Cognitive Communications and Networking*, vol. 3, no. 4, pp. 563-575, Dec. 2017.

Submitted by: Roll No. 623160 & Roll No. 623192

National Institute of Technology Andhra Pradesh | ECE Department | November 2026